

Evan J. Turner

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EDUCATION

University of Wisconsin-Madison, Madison, WI

M.S., Mechanical Engineering | GPA: 3.89

Expected Dec 2026

Clarkson University, Potsdam, NY

B.S. with Honors, Mechanical Engineering (Minor: Biomedical Engineering) | GPA: 3.76

May 2024

Leadership: President & Captain, Men's Club Ice Hockey Program

RELEVANT EXPERIENCE

Graduate Research Assistant, Cardiovascular Fluid Dynamics Lab (UW-Madison)

Aug 2024 – Present

- Building an image processing and simulation pipeline for evaluating wall shear stress metrics' ability to identify heightened risk pregnant patient groups using doppler US-derived low-fidelity computational fluid dynamics (CFD) for N = 82 subjects.
- Modeled patient-specific Fontan hemodynamics via physiologic CFD to predict energy loss and flow distribution to aid surgical planning for a time-sensitive clinical case.
- Characterizing nephrostomy tube blockage from exchange procedures by designing an experimental setup that established quantitative diagnostic metrics.
- Analyzed urethral biomechanics in stricture patients by developing a non-invasive MRI and CFD framework.

Production Engineer (Part Time), Calimetrix LLC

Aug 2026 – Present

- Assessing manufacturing systems to design, test, and integrate equipment upgrades that scale production efficiency.
- Designing housing units in SolidWorks for new devices, transitioning prototypes into scalable injection molded products.

Biomedical Engineering Co-op, ZOLL Medical Corporation

Jan 2023 – Aug 2023

- Developed novel polymer fabrication process, increasing process reliability from <5% to almost 100%, boosting fabrication speed by ~40x and device sensitivity by 10x.
- Prototyped mechanical test fixtures by modeling systems in SolidWorks and manufacturing components via 3D printing and laser cutting.
- Designed and executed engineering studies to verify new product performance using custom-built MATLAB signal processing tools for time-frequency data analysis.

Research Assistant, Connective Tissue Mechanics Lab (Clarkson University)

May 2021 – May 2024

- Developed an ex-vivo model to quantify post-retraction intradiscal injection leakage, establishing a benchmark for rapid risk screening. Led a study using this technique to investigate post-procedure movement as a risk factor.
- Designed a low-cost, wearable spinal motion tracking system utilized across multiple biomechanics studies, including investigation of spinal motion coordination post-fatigue in competitive Nordic skiers.

PUBLICATIONS & PRESENTATIONS

- **Journal Article:** "Establishing a Through-Puncture Model for Assessing Post-Injection Leakage", European Spine Journal, 2022.
- **Conference Presentation:** "WSS-Based Differentiation of Hypertensive Disorders of Pregnancy Using US-Based CFD", ASME SBC, 2025. (*Finalist, MS Student Paper Competition*)
- **Journal Article:** "Biomechanical Analysis of Lumbar Coordination in Nordic Skiers During Double Poling", Communications in Kinesiology, 2026.
- **Journal Article:** "Non-Invasive Methods for Bladder Biomechanics During Voiding Using Uro-Dynamic MRI", Bioengineering, 2026.
- **Conference Presentation:** "MRI-Based CFD in Urethral Stricture Biomechanics", ISMRM Annual Meeting, 2025.

TECHNICAL SKILLS

- **Computer Aided Design:** SolidWorks (CSWA for Mechanical Design), AutoCAD, Fusion360, KiCAD
- **Programming & Simulation:** MATLAB, Python, C++, LabVIEW, ANSYS, Altair, SimVascular
- **Manufacturing & Fabrication:** 3D Printing (FDM/SLA), Laser Cutting, CNC Routing, Lathe, Soldering